

Cooperative LiDAR-Aided ~~Augmented~~ Self-Localization of CAVs ^{*}

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Abstract. In the rapidly advancing realm of Connected Autonomous Vehicles (CAVs), achieving reliable and precise positioning is paramount. This paper presents a comprehensive approach integrating vehicular sensing, communication, and advanced filtering techniques to enhance vehicle positioning in urban areas. By leveraging LiDAR point clouds along with a light and accurate object detector, we create cohesive environmental sensing that improves situational awareness in autonomous systems. Central to our methodology is the integration of the Labeled Multi-Bernoulli Mixture (LMBM) tracking filter, which offers a probabilistic framework for dynamic state estimation in environments characterized by high uncertainty and clutter. In turn, enhanced object locations are exploited as anchors for vehicular self-localization via an Extended Kalman filter (EKF). Our experimental results demonstrate that ~~integrating~~ the proposed solution significantly enhances vehicular positioning accuracy.

Keywords: Cooperative sensing · Kalman filtering · Bernoulli tracking.

(*) That cannot be tolerated in critical safety-related services

1 Introduction

Global Navigation Satellite Systems (GNSSs) have widespread application in Connected Autonomous Vehicle (CAV) scenarios. However, conventional GNSS systems suffer from multipath propagation in urban environments, leading to high ~~potent~~ positioning errors [6]. A promising solution to overcome this problem is represented by ~~implicit~~ cooperative positioning (ICP) [2], ~~and sensing systems~~ that leverage the presence of multiple actors in the same driving area to enhance positioning accuracy thanks to the coherent fusion of localization measurements acquired by nearby vehicles, ~~such as~~ ^{such as}

This paper introduces a novel method for cooperative LiDAR-based vehicular self-positioning that leverages data sharing among ~~multiple~~ vehicles to enhance the localization accuracy ~~of vehicles~~. The innovative aspect of this method lies in

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its ability to address and mitigate key challenges such as false detections, object-measurement association, and measurement-measurement association which are typically ignored by conventional methods. By integrating LiDAR data from various vehicles, the method ensures a more reliable and accurate vehicle positioning. Validation was conducted using a real-world dataset where multiple vehicles approached a zebra crossing on a two-way urban road featuring stationary road elements at the crossing. The results demonstrate the method's effectiveness in improving positional accuracy and robustness in complex urban environments, underscoring its potential for practical applications in autonomous driving and advanced driver-assistance systems.

The rest of the paper is organized as follows. In section 2, the cooperative sensing-based self-positioning solution is presented. Section 3 presents the details of the experiment and provides a quantitative assessment, followed by concluding remarks in section 4.

2 Cooperative Self-Positioning

The proposed method fundamentals on implicit cooperative sensing and positioning. Vehicles equipped with LiDAR sensors concurrently generate point clouds. These point clouds encompass useful information for autonomous driving such as the location of road elements, walls and road's slope. In this end-to-end method, the object detector proposed in Section 2.1 clusters separable points comprised of 3D in the vehicle body reference system and identifies them as objects. Among these detected objects, ~~there exist~~ noise-derived detections. Along with the LiDAR sensor, ~~the other two onboard units~~ (GNSS and Inertial Measurement Unit (IMU)) are actively engaged. Here, the GNSS positioning is enhanced with a Satellite-Based Augmentation System (SBAS), ~~if~~ ^{where} available, which requires LOS conditions with at least two geosatellites. Otherwise, Single-Point Positioning (SPP) is considered. Additionally, vehicular heading is required for LiDAR-GNSS calibration. Some approaches used for the calculation of vehicular heading are visual odometry, dual-antenna GNSS localization, and fusion of GNSS and IMU [3]. For the sake of simplicity, we used single-antenna GNSS fixes to calculate vehicle heading and then calibrated the rotational component of LiDAR and GNSS reference frames. ~~As of here, these operations are done on every vehicle in close proximity if decentralized cooperation applies, or which are connected to the same Road Side Unit (RSU) for centralized cooperation.~~ While the proposed method can be utilized either way, this paper will focus on centralized cooperation. In turn, the following data is exchanged with RSU via Vehicle-to-Infrastructure (V2I) communications: detected object locations in the local framework and GNSS position fix. RSU performs a series of operations. Firstly, gathered data ~~is~~ ^{are} synchronized and detected objects by different vehicles are projected onto the global reference system. At this step, a Multi-Target Tracking (MTT) filter tracks the static road elements in the presence of clutters and high uncertainty. Here, vehicles are considered different sensors making observations in the same surveillance area and a special form of Labeled

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Multi-Bernoulli Mixture (LMBM), also known as multi-sensor LMBM, is utilized. This operation returns object locations with lower uncertainty to be taken into account as anchors for the subsequent steps. Worth noting that, the accuracy and precision of target locations in the global reference frame are tightly correlated to the number of vehicles as the higher the number of vehicles, the better the accuracy. These objects are back-propagated to each vehicle via V2I communications. Within this transmission, GNSS position fix, detected targets with clutters, and augmented object locations (social information) are available at each vehicle. Subsequently, vehicles individually employ a custom Extended Kalman Filter (EKF) given in section 2.3 and leverage their global positioning.

2.1 Grid-based Spatial Detector

The Grid-Based Spatial Detector (GSD) employed in this work is designed to isolate sets of points in 3D space that ~~are reflected~~ ^{have been generated} by objects. The process is initiated by inserting a point cluster at a known position within the point cloud to address segmentation noise. Subsequently, the point cloud is projected onto a 2D space and segmented to generate a density map. This density map is ~~subsequently~~ ^{finally} filtered to reduce noise-induced segments. At the core of this object detector is a density-based clustering method known as density-based spatial clustering of application with noise (DBSCAN) ^(ref) utilized to generate clusters in the density map. In turn, the inserted point cloud has been recognized among all detections and the segmentation bias has been eliminated by shifting clusters. Consequently, clusters with small dimensions are averaged and represented as single-point objects, ensuring accurate and reliable object detection.

2.2 Cooperative LMBM

MTT is a critical process ^{performed} in autonomous systems, ~~involving~~ ^{for} detecting and tracking multiple objects ~~simultaneously~~ in a dynamic environment. ~~One~~ ^{Am} advanced method ~~used in MTT is the~~ LMBM [4], ^{which efficiently handles} measurement-target data association and clutter. The process? Model? Bernoulli birth model that initiates ~~Bernoullis~~ at predefined locations.

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The projection of ~~detected objects by vehicles~~ on a global reference system depicts a multi-sensor configuration at a surveillance area with independent imaging sensors. Likewise, the detected objects by these sensors are independent. Therefore, Bernoulli's are independent and follow separate Bayesian recursions. A Bernoulli Random Finite Set (RFS) β is comprised of components at time t , $RFS_{\beta} \triangleq \{\{T, L\}_{\beta}, r_{e,t,\beta}, \mu_{t,\beta}, \Sigma_{t,\beta}\}$, where T_{β} is the birth time, L_{β} is a unique label assigned to Bernoulli's born at time t , and $\mu_{t,\beta}$ is the first central moment and $\Sigma_{t,\beta}$ is the covariance of Gaussian distribution. Bernoullis follow a static motion model, namely $\mu_{t,\beta} = \mu_{t-1,\beta}$, knowing the existence of fixed ~~targets~~ ^{targets?} in the considered driving scenario.

For the propagation of existence probability, r_e , and marginal association probabilities over time, Gibbs's sampling [5] has been employed thanks to its capability to handle high dimensional problems. After deploying steps gating,

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pruning and capping, target states are obtained by processing the remainder Bernoullis. These states contain the spatial attribute of objects in 2D, denoted by object set $\mathcal{O} = \{\mathbf{x}_{o,t}\}$, and their corresponding covariance to be used by the filter introduced in the following section.

2.3 System Model

A set of vehicles $\mathcal{V} = \{1, \dots, N_V\}$ cruise in a 2D area. Each vehicle v , with $\mathcal{V}$, is represented by its state $\mathbf{x}_{v,t} = [\mathbf{u}_{v,t}, \mathbf{v}_{v,t}]$ where $\mathbf{u}_{v,t}$ and $\mathbf{v}_{v,t}$ are the position and 2D velocity, respectively. We assume that the vehicle's state evolves over time according to a nearly constant velocity motion model, defined

EKF is employed for tracking as follows. States are predicted ...

$$\mathbf{x}_{v,t} = \mathbf{F}\mathbf{x}_{v,t-1} + \mathbf{L}\mathbf{q}_{v,t-1}, \quad (1)$$

where $\mathbf{F} = [\mathbf{I}_2 T_s \mathbf{I}_2; \mathbf{0}_{2 \times 2} \mathbf{I}_2]$ and $\mathbf{L} = [0.5T_s^2 \mathbf{I}_2; T_s \mathbf{I}_2]$, T_s the sampling time, while $\mathbf{q}_{v,t-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_2 \sigma_{v,U}^2)$ is a zero-mean Gaussian random variable with standard deviation $\sigma_{v,U}$ modeling the acceleration uncertainty of the v -th vehicle.

The predicted states according to the motion model are exchanged with RSU along with detected objects recognized by GSD given in section 2.1. Subsequently, the LMBM filter given in section 2.2 is utilized. Then the leveraged object locations are back-propagated to vehicles and object-object association between detected objects and leveraged targets is established according to maximum likelihood. association? (ref)

For the update step of the EKF, both GNSS and object-derived measurements are utilized to enhance vehicular positioning. The GNSS measurement of vehicle v can be modelled as follows:

are

$$\rho_t^v = \mathbf{T}\mathbf{x}_{v,t} + \mathbf{n}_{v,t}, \quad (2)$$

where $\mathbf{T} = [\mathbf{I}_2, \mathbf{0}_{2 \times 2}]$, while $\mathbf{n}_{v,t} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_2 \sigma_{v,P}^2)$ is a zero-mean Gaussian random variable with standard deviation $\sigma_{v,P}$ characterizing the GNSS positioning accuracy of the v -th vehicle.

An object $o \in \mathcal{O}$ with state $\mathbf{x}_{o,t} = [\mathbf{u}_{o,t}^x, \mathbf{u}_{o,t}^y]$ provided by RSU has more precise positioning with respect to the detection j , such that $\mathbf{x}_{j,t} = [\mathbf{u}_{j,t}^x, \mathbf{u}_{j,t}^y]$, generated by the same object on vehicle v . Leveraging this fact, so o and j , the measurement models $\rho_r^{(j,v)}$, distance between an object j and vehicle v , and $\rho_\alpha^{(j,v)}$, clockwise angle from positive x-direction between an object j and vehicle v are defined as

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$$\rho_{r,t}^{(j,v)} = \rho_t^v - \mathbf{x}_{j,t} + \mathbf{n}_{r,t}^{j,v}, \quad (3)$$

$$\rho_{\alpha,t}^{(j,v)} = \angle(\rho_t^v, \mathbf{x}_{j,t}) + \mathbf{n}_{\alpha,t}^{j,v}, \quad (4)$$

← where $\angle$ is the angle in radians while $\mathbf{n}_{r,t}^{j,v} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_2 \sigma_r^2)$ is a zero-mean Gaussian random variable with standard deviation σ_r and $\mathbf{n}_{\alpha,t}^{j,v} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_2 \sigma_\alpha^2)$ is a zero-mean Gaussian random variable with standard deviation σ_α . These measurements are used to improve the vehicle state by range $\rho_{r,t}^{(o,v)}$ and angle $\rho_{\alpha,t}^{(o,v)}$ information derived from enhanced object o and vehicle v .

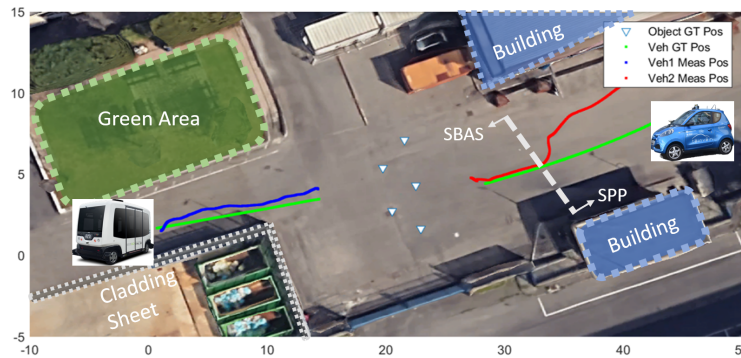


Fig. 1. Trajectories show GNSS measurements along with RTK-corrected ground truth (GT) positions, SBAS and SPP available regions along with prototypes: vehicle one (right) and vehicle two (left)

Data collected by sensors were aggregated and processed at a central computational unit offline. As concerns the computational cost,

ation and Results

In order to evaluate the performance of the proposed method, we collected real-world data experiments using two different instrumented prototype vehicles, a tricycle with custom sensors and a prototype bus as shown in Fig. 1. Collected data includes SBAS measurements (5 cm accuracy), uncorrected GNSS measurements, and LiDAR point clouds. Five objects detectable by LiDAR were used to enhance positioning, aiming to evaluate localization improvement compared to ICP. The setup, shown in Fig. 1, featured both vehicles converging towards an area with the deployed objects. For further elaboration on the experimental campaign and dataset, refer to [1].

During the implementation of the proposed method, an assumption of lossless and zero-latency V2I communication has been made. After the generation of sensory data, all information was gathered at a central computational unit (computer) and by doing so, vehicular communication has been realized. An important keynote for the computational cost of this algorithm is that the bottleneck operation is LMBM in which the number of hypotheses can increase exponentially. While this problem could be overcome by investigating lighter MTT algorithms, we limited the number of hypotheses and Gibbs's samples and initiated Bernoullis in the centre of the surveillance area, so near to objects.

Fig. 2 demonstrates the Empirical Cumulative Density Function (ECDF) of vehicle positioning error considering ICP with and without multipath compensation, raw GNSS ego vehicle position, proposed method, referred to as Cooperative Self-Positioning (CSP), along with Circular Error Probability with 95% confidence (CEP95). When CEP95 is concerned, CSP achieved positioning improvement of 68% and 66% compared to GNSS for veh1 and veh2 respectively. Besides, CSP shows similar performance to the benchmark method ICP which was built on the assumptions that make ICP unemployable in a real application.

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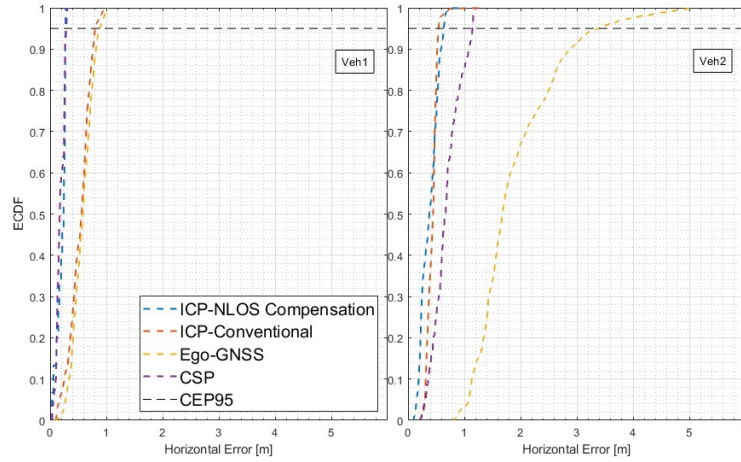


Fig. 2. Performance comparison of CSP against ICP with and without NLOS compensation and Ego-GNSS.

4 Conclusion

This paper proposed a cooperative sensing-aided positioning method relying on environmental perception with LiDAR sensors. This model-based algorithm handles the uncertainty of a dynamic environment by exploiting collaborative sensing and employing the LMBM filter. This method has been assessed in a real-world application and its effectiveness has been reported. The planned future work is to establish V2I communications and integrate RSU as both an observer and the central computational unit.

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